

# Priority-Aware Encoding for Bandwidth-Efficient Real-Time Classification in 5G Networks

Chengzhang Li<sup>†</sup> Peizhong Ju<sup>‡</sup> Atilla Eryilmaz<sup>†</sup> Ness B. Shroff<sup>†</sup>

<sup>†</sup>Department of Electrical and Computer Engineering, The Ohio State University, Columbus, OH, USA

<sup>‡</sup>Department of Computer Science, University of Kentucky, Lexington, KY, USA

**Abstract**—The growing demand for real-time edge intelligence in 5G networks requires communication systems that are both bandwidth-efficient and resilient to transmission errors. Prior work has proposed compression schemes for edge classification tasks, but these often rely on oversimplified transmission models that ignore practical constraints of 5G systems, such as LDPC-coded block transmissions and block error rates (BLER). In this paper, we address this gap by integrating a realistic 5G uplink transmission model into the design of edge classification systems. Specifically, we propose a novel source encoding framework based on Priority Encoding Transmission (PET), which enables progressive, priority-aware recovery of compressed image features under probabilistic code block loss. PET operates entirely at the information bit level, making it compatible with existing 5G physical layer standard. In particular, we (i) formulate an optimization problem that determines the optimal PET encoding scheme to maximize the expected classification accuracy; (ii) propose a computation procedure to find a theoretical upper bound for the expected classification accuracy; (iii) gain insights and theoretical results for this problem; (iv) and develop a low-complexity algorithm to efficiently find a two-priority-level PET configurations. Extensive simulations demonstrate that our proposed method is near-optimal and significantly outperforms our baselines.

## I. INTRODUCTION

The rapid proliferation of intelligent Internet of Things (IoT) applications—such as real-time video analytics, autonomous vehicles, and industrial automation—has significantly increased the demand for low-latency, high-reliability communication at the network edge [1]–[3]. In such systems, low-cost edge devices equipped with sensors must frequently offload computation-intensive tasks, such as image classification or object detection, to nearby edge servers. Due to both the limited processing capabilities of low-cost edge devices and the bandwidth constraints of wireless networks (especially when supporting a large number of concurrent transmissions), it is crucial for these devices to compress their data before transmission. This not only reduces bandwidth usage but also enables timely processing at the server side using the compressed representations.

There are many technologies for data compression at network edge in the literature [4]–[6]. Among them, [6] recently proposed a single-codebook split computing framework to handle time-varying channel conditions, in which a single compression NN is used across different channel conditions. The number of transmitted neurons is dynamically adjusted according to the available data rate.

A major limitation of [6] lies in its simplified transmission model, which does not reflect any real-world communication protocols such as 5G NR [7]. In particular, the model overlooks key physical layer constraints such as error-prone code blocks, block error rates (BLER), low-density parity check (LDPC) coding structure [8], and the real-time, no-retransmission nature of 5G's ultra-reliable low-latency communication (URLLC) mode [9]. As a result, the proposed scheme may not perform as expected in practical deployments.

In this paper, we address this limitation by considering the 5G uplink transmission process, where compressed data is sent over LDPC-coded physical uplink shared channels (PUSCH) under tight latency constraints. Specifically, we consider the URLLC setting, where all data must be delivered within a single transmission time interval (TTI), and retransmissions are not allowed. This more accurately reflects the operational constraints of many real-time 5G applications, such as autonomous driving, industrial control, and augmented reality.

While designing a 5G-compliant solution for real-time edge classification, we observe that integrating error-resilient source coding is beneficial for maintaining classification accuracy in the presence of block errors. Among the many error-resilient coding techniques available, we identify Priority Encoding Transmission (PET) [10] as particularly well-suited for integration with the single-codebook NN compression design proposed in [6]. Unlike many erasure coding schemes that treat all bits equally, PET supports progressive decoding—allowing higher-priority data to be recovered first—which aligns naturally with the observation that the top neurons in the output layer of a single-codebook compression NN are more critical for accurate classification.

Our goal is to develop a PET encoding scheme to maximize the expected classification accuracy. To that end, we propose a computation procedure to find a theoretical upper bound for the objective. A key insight of our study is that, despite the complexity of the problem, an optimal or near-optimal PET solution can often be achieved with just two priority levels. We prove this formally in the case where the accuracy function is linear, and provide a theoretical guarantee when the accuracy function is concave.

Motivated by this insight, we develop an efficient and low-complexity algorithm that searches over all feasible two-priority-level PET configurations and selects the one that maximizes the expected accuracy. This approach has a very low time complexity, making it suitable for deployment in latency-

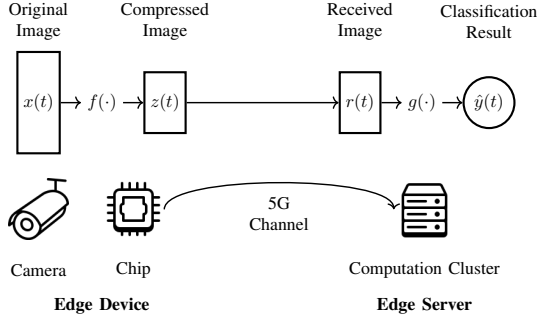


Figure 1: Model for image classification at network edge.

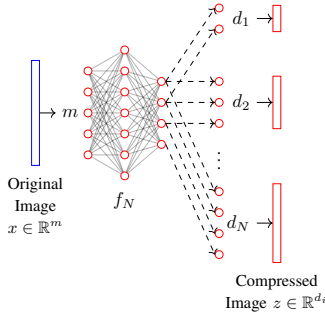


Figure 2: Single codebook design for compressor  $f(\cdot)$ .

sensitive 5G environments. Our numerical results show that the proposed two-priority-level PET solution achieves significant accuracy gains over naive transmission strategies that do not employ source encoding.

The main contributions of our work are:

- We propose a novel PET-based source encoding framework tailored for real-time edge classification tasks over 5G, explicitly accounting for practical constraints such as LDPC-coded transmissions, code block errors, and the no-retransmission requirement in URLLC scenarios.
- We develop a computation procedure to find a theoretical upper bound of the objective. For a slightly relaxed problem, we show that two priority levels are sufficient for optimality when the accuracy function is linear, and provide a theoretical guarantee for 2-level solutions when the accuracy function is concave.
- We develop a low-complexity algorithm that can find the optimal 2-level PET configurations. This algorithm is designed to operate within the stringent timing requirements of 5G, making it suitable for deployment in latency-sensitive applications.
- We evaluate the performance of our proposed scheme using both synthetic settings and real neural network models (trained on CIFAR-10), demonstrating that the 2-level PET solution achieves near-optimal performance and significantly outperforms naive baselines without source encoding.

## II. SYSTEM MODEL AND PROBLEM STATEMENT

### A. Image Compression and Classification

Consider an edge classification model illustrated in Fig. 1. An edge device captures images from its environment and classifies them into one of  $L$  categories, where  $L$  represents the number of classes. Due to the limited computational capacity of its integrated low-cost chip, the device offloads images to a compute server with a high-performance computing cluster for classification. Due to bandwidth constraints, the device compresses the image before transmitting it to the edge server.

As introduced in [6], to enable efficient data transmission, a single neural network is employed for compression across multiple output dimensions. As shown in Fig. 2, the compression function  $f(\cdot)$  consists of a single neural network,  $f_N$ , with an output dimension of  $d_N$ . At each time slot  $t$ , the edge device utilizes  $f_N$  to compress data and selects the top  $d(t)$  neurons for transmission to the server. Here  $d(t) \in \{d_1, d_2, \dots, d_N\}$  is the size of the compressed image  $z(t)$  in Fig. 2. In this paper, we refer to the information contained in a neuron as a “word.” Each word consists of  $B$  bits of information, where  $B = 32$  for single precision and  $B = 64$  for double precision.

In Fig. 1, the classifier  $g(\cdot)$  at the server is able to handle each input dimension  $d_1, d_2, \dots, d_N$  and provide a classification result. Note that the expected classification accuracy depends on the size of the input image, and we assume a larger image can give a higher classification accuracy. Let  $\alpha(d_i)$  denote the expected classification accuracy of  $g(\cdot)$  for an input image with dimension  $d_i$ . We assume  $\alpha(d_1) \leq \alpha(d_2) \leq \dots \leq \alpha(d_N)$ .

We can see a trade-off between communication bandwidth and classification accuracy. That is, if the image is compressed to a smaller size, the channel bandwidth required to transmit this image will be smaller, but the expected classification accuracy for this image is also lower, and vice versa. In this paper, we will study the case when the channel bandwidth is limited, and our goal is to maximize the expected classification accuracy.

### B. 5G Transmission and Source Encoding

As discussed in Section I, a major limitation of the model used in [6] is its reliance on a simplified transmission abstraction that does not reflect real-world communication protocols. In this paper, we consider a more realistic 5G transmission model, where compressed image data is transmitted from the edge device to the server over the physical uplink shared channel (PUSCH). The system operates under a strict real-time constraint, wherein each image must be delivered within a single TTI, which can range from 0.0625 ms to 1 ms depending on the numerology. This setup comes from URLLC scenarios in 5G, where the end-to-end latency requirement is typically less than 1 ms [9]. Under this constraint, retransmissions in subsequent TTIs are not permitted, as they would violate the latency requirement. Therefore, the receiver must perform classification based solely on the data received within a single TTI, even in the presence of code block decoding failures.

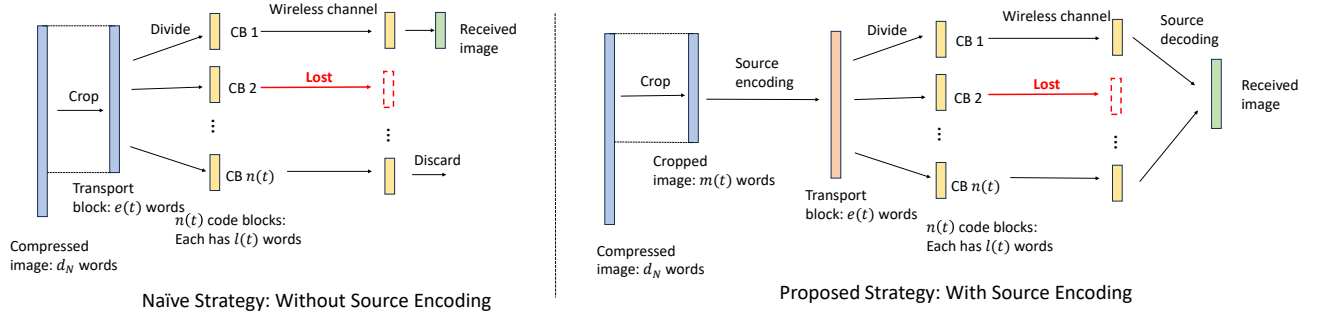


Figure 3: A motivating example for source encoding.

We now briefly introduce the relevant aspects of physical layer coding in 5G NR. In each TTI, the Medium Access Control (MAC) layer organizes data into a transport block, which is then forwarded to the physical layer for transmission. For PUSCH, 5G employs LDPC coding for channel encoding. The maximum size of each LDPC code block depends on the selected base graph: it is 8,848 bits for Base Graph 1 and 3,840 bits for Base Graph 2 [11]. When the transport block size exceeds the maximum allowable code block size, the transport block is segmented into multiple code blocks, each of which is encoded and decoded independently. This segmentation introduces vulnerability to partial packet loss: failure to decode any code block may lead to incomplete reception of the transmitted data.

The design of the 5G scheduling algorithm is outside the scope of this paper. Instead, we assume a given 5G scheduling algorithm that determines resource block (RB) allocation and modulation and coding scheme (MCS) selection. For each TTI  $t$ , the scheduling outcome and 5G configurations determine the number of code blocks and their respective sizes. We consider a device. Let  $n(t)$  denote the number of code blocks,  $l(t)$  denote the size of each code block, and  $e(t) = n(t)l(t)$  denote the total transport block size. For simplicity, we normalize  $l(t)$  in terms of “words.” For example, for a code block of 3,840 bits and  $B = 64$  bits per word, we have  $l(t) = 3840/64 = 60$ . Here  $e(t) = n(t)l(t)$  can be considered as the bandwidth allocated to the device at TTI  $t$ .

Given this setup, the problem we aim to address is: for a compressed image  $z(t)$ , and given the transmission parameters  $n(t)$  and  $l(t)$ , how should the edge device transmit  $z(t)$  to maximize the expected classification accuracy at the server?

We use *source encoding* to solve this problem. Fig. 3 illustrates a motivating example for the use of source encoding. In this scenario, the size of the compressed image exceeds the available transport block size, i.e.,  $d_N > e(t)$ . A naïve approach (the left part of Fig. 3), without any source encoding, is to simply crop the compressed image to the first  $e(t)$  words, which are then divided into  $n(t)$  code blocks for transmission. However, this approach is highly sensitive to code block errors, especially when errors occur in the early blocks. In 5G systems, the BLER is typically around 10%, but can rise to 30% under poor channel conditions, depending on the link

adaptation and scheduling algorithms used [12], [13]. Given our assumption that retransmissions are not allowed, the server must perform classification based solely on the data it receives in a single TTI. The classifier expects the top- $d$  words (where  $d \in d_1, \dots, d_N$ ) of the compressed image as input. If a code block is lost—e.g., the second code block fails to decode—the classifier cannot utilize the subsequent blocks ( $3, 4, \dots, n(t)$ ), and must rely solely on the information in the first code block. This leads to a substantial degradation in classification accuracy.

In contrast, as shown in the right part of Fig. 3, employing a source encoding algorithm with redundancy leads to a more robust outcome. Instead of simply cropping the compressed image, we first extract the top  $m$  words (where  $m \leq e(t)$ ) and apply a redundancy-aware source encoding scheme to generate a transport block of size  $e(t)$ . This redundancy allows the server to recover a greater portion of the original data, even in the presence of code block errors, thereby improving classification accuracy.

Numerous error-resilient redundancy-aware coding schemes have been proposed in the literature, including LT codes [14], Raptor codes [15], and unequal error protection (UEP) strategies [16]. These techniques are widely adopted in communication systems to enhance robustness against data loss and corruption.

Among these, Priority Encoding Transmission (PET) [10] is particularly well-suited for our 5G classification setting. Unlike other coding schemes that treat all data uniformly, PET enables progressive decoding of prioritized data segments, which aligns naturally with our model, where the top- $d$  words of the compressed image are the most critical for classification. In the following section, we provide a brief introduction to PET and describe how it can be integrated into our system.

### C. PET Encoding

PET is an error-resilient source encoding scheme designed to prioritize the recovery of critical information in unreliable transmission environments [10]. Unlike traditional erasure coding methods which treat all data segments equally, PET assigns varying levels of importance to different parts of the data. This allows higher-priority information to be recovered earlier and more reliably, even in the presence of partial data loss.

Formally, PET encodes a message consisting of  $m$  words into  $n$  packets, each containing  $l$  words. Each word  $i \in 1, 2, \dots, m$  is assigned a priority value  $\rho_i \in (0, 1]$ , such that  $\rho_1 \leq \rho_2 \leq \dots \leq \rho_m$ . The PET encoding guarantees that, for any such message, the  $i$ -th word can be decoded from any subset of packets covering at least a  $\rho_i$  fraction of the total  $n$  packets [10]. This progressive property makes PET well-suited to real-time, loss-tolerant systems.

In the context of transmitting compressed images for edge classification, PET provides a natural and effective solution. As shown in Fig. 2, the output of the compressor consists of a sequence of neurons, where earlier neurons (e.g., the first few) typically carry more important features for classification. By assigning priorities accordingly—i.e., lower  $\rho_i$  for higher-priority neurons—PET enables the receiver to progressively recover the most critical information as more code blocks are successfully received.

This design allows the system to combat code block errors: even if some code blocks are lost, the classifier can still make use of the highest-priority neurons that have been successfully decoded. Hence, PET significantly enhances the robustness of inference in 5G edge scenarios, maintaining classification accuracy even under challenging channel conditions.

#### D. Problem Statement

In this paper, we study the problem of designing and optimizing a PET scheme for compressed images in a 5G-based classification system. Given a 5G scheduling result, our goal is to maximize the expected classification accuracy by optimally selecting PET encoding parameters under lossy transmission conditions.

For each TTI  $t$ , we assume the following system parameters are known:

- $d_N$ : The size of the compressed image.
- $n(t)$ : The number of code blocks allocated for transmission.
- $l(t)$ : The size of each code block.
- $p(t)$ : The block error rate, assumed to be identical and independent<sup>1</sup> across all blocks within TTI  $t$ .

For ease of presentation, we use “ $d$ ” and “ $d_N$ ” interchangeably in the remaining parts of this paper.

Let  $\eta(t)$  denote the number of code blocks successfully received (out of  $n(t)$  transmitted blocks) during TTI  $t$ . We assume that the distribution of  $\eta(t)$  follows a binomial model<sup>2</sup>:

$$p_k = \mathbb{P}(\eta(t) = k) = \binom{n(t)}{k} p(t)^k (1 - p(t))^{n(t)-k}, \quad k \leq n(t).$$

We also assume that we know the *expected accuracy function* of the classifier  $g(\cdot)$ , denoted by  $\alpha(x)$ , which represents the classification accuracy when the input dimension is  $x$

( $1 \leq x \leq d_N$ ). This  $\alpha(\cdot)$  can be obtained by testing the classifier  $g(\cdot)$  with input dimensions. We assume  $\alpha(x)$  monotonically increases with  $x$ .

Under a PET encoding scheme with priority  $\rho_1, \rho_2, \dots, \rho_m$ , when receiving  $k$  code blocks ( $k = 0, 1, 2, \dots, n(t)$ ), the size of the received (reconstructed) image after PET is

$$\delta(k) = \max\{i : \rho_i \leq k/n(t)\}. \quad (1)$$

This ensures that only words with priority levels up to  $k/n(t)$  are recoverable. And the expected classification accuracy at TTI  $t$ , denoted by  $J(t)$ , is given by

$$J(t) = \sum_{k=0}^{n(t)} p_k \alpha(\delta(k)). \quad (2)$$

The problem we study is that, under given  $d(t)$ ,  $n(t)$ ,  $l(t)$ ,  $p(t)$ , and  $\alpha(\cdot)$ , how to find the optimal PET encoding scheme to maximize  $J(t)$ . In particular, we need to

- Determine the optimal message length  $m^*$ .
- Find the optimal priorities under PET:  $\rho_1^*, \rho_2^*, \dots, \rho_{m^*}^*$ .
- Construct the PET encoding solution with  $\rho_1^*, \rho_2^*, \dots, \rho_{m^*}^*$ .

By solving this problem, we ensure that the classifier receives the most informative portion of the compressed image under varying transmission conditions, thereby maximizing classification accuracy while adapting to 5G network dynamics.

This optimization needs to be performed within a single TTI in the 5G system, which imposes stringent timing constraints. Since a TTI can be shorter than 1 ms, the PET encoding scheme needs to be computationally efficient to ensure that the encoding, transmission, and subsequent classification occur within the real-time deadline. This stringent timing requirement means that we need to come up with an extremely efficient way of solving this problem.

Importantly, PET can be seamlessly integrated into existing 5G systems without requiring any changes to the physical layer. PET operates entirely at a higher layer, modifying only the information bits in the transport blocks. The encoded data is then passed to the physical layers, which handle scheduling and LDPC coding as usual. This allows PET to be layered on top of existing 5G protocols (such as LDPC-coded PUSCH in the uplink), ensuring compatibility with current standards and hardware.

### III. PERFORMANCE BOUND

In this section, we derive a theoretical upper bound for the objective  $J$  in (2). This upper bound serves two purposes: (i) it provides a performance benchmark for evaluating our proposed algorithm, and (ii) the insights gained during its derivation offer useful guidance for our algorithm design in the next section.

#### A. Formulating An Optimization Problem

Based on the discussion in [10], the girth of a PET system is defined as

$$g = \sum_{i=1}^m \frac{1}{\rho_i}. \quad (3)$$

<sup>1</sup>This block error rate can be controlled and estimated by link adaption algorithms, e.g., [12], [13].

<sup>2</sup>Even if  $\eta(t)$  does not follow a binomial distribution, all results in this paper remain valid as long as the distribution of  $p_k$  is known. Nevertheless, the binomial model serves as a good example scenario due to its wide applicability and analytical characteristics.

For any PET system, we have  $e = nl \geq g$  [10]. That is, the PET system we design must satisfy the following constraint:

$$\sum_{i=1}^m \frac{1}{\rho_i} \leq nl. \quad (4)$$

We can formulate the following optimization problem:

$$\begin{aligned} \text{OPT: } \max_{m, \rho'_i s} \quad & H = \sum_{k=0}^n p_k \alpha(\delta(k)), \\ \text{s.t.} \quad & \delta(k) = \max\{i : \rho_i \leq k/n\}, \quad k = 1, 2, \dots, n, \\ & \delta(0) = 0, \\ & \sum_{i=1}^m \frac{1}{\rho_i} \leq nl, \\ & 1 \leq m \leq d, m \in \mathbb{N}, \\ & 0 < \rho_i \leq 1, i = 1, 2, \dots, m. \end{aligned}$$

Let  $H^*$  denote the optimal objective of OPT. We have  $H^* \geq J^*$ . That is,  $H^*$  is an upper bound of  $J$ .

It is clear that when  $d \leq l$  (i.e., the size of the compressed message is smaller than the size of one code block), the optimal solution of OPT is  $m^* = d$  and  $\rho_i^* = 1$  for all  $i = 1, 2, \dots, d$ . This means we put the entire compressed image into each code block, and the server can decode the entire image if it successfully receives any one code block. This case is trivial and not the main focus of this paper. Therefore, in the remaining part of this paper, unless we have any specification, we assume  $d > l$ .

### B. Re-formulating and Relaxing OPT

Regarding the optimal solution to OPT, we introduce the follow proposition.

**Proposition 1** *There exist an optimal solution  $m^*$  and  $\rho_1^*, \rho_2^*, \dots, \rho_{m^*}^*$  to OPT such that for each  $i = 1, 2, \dots, m^*$ , we have  $\rho_i^* = q_i/n$ , where  $q_i$  is an integer.*

**A Proof Sketch** We prove Proposition 1 by construction. Suppose  $m'$  and  $\rho'_1, \rho'_2, \dots, \rho'_{m'}$  is a set of optimal solution to OPT. Then we construct  $m^* = m'$  and  $\rho_i^* = \lceil \rho'_i n \rceil / n$  for each  $i = 1, 2, \dots, m'$ , where  $\lceil \cdot \rceil$  is the ceiling function. It is easy to verify  $m^*$  and  $\rho_1^*, \rho_2^*, \dots, \rho_{m^*}^*$  is a feasible and optimal solution to OPT. ■

Proposition 1 tells us in an optimal solution to OPT, the value of  $\rho_i$  can only be  $\frac{1}{n}, \frac{2}{n}, \dots, \frac{n-1}{n}$ , or 1. And we can utilize this to re-formulate OPT.

Let  $x_j \in \mathbb{N}$  denote the number of  $i$ 's that makes  $\rho_i = j/n$ . Then we can re-formulate OPT as:

$$\begin{aligned} \text{OPT-}x: \max_{x'_j s} \quad & H = \sum_{k=0}^n p_k \alpha\left(\sum_{j=1}^k x_j\right), \\ \text{s.t.} \quad & \sum_{j=1}^n \frac{x_j}{j} \leq l, \\ & \sum_{j=1}^n x_j \leq d, \\ & x_j \geq 0, x_j \in \mathbb{N}, j = 1, 2, \dots, n. \end{aligned}$$

Based on Proposition 1, it can be shown that OPT- $x$  is equivalent to OPT.

Note that in OPT, the accuracy function  $\alpha(x)$  is defined for integer  $x$ , which is not easy to solve. Therefore, we relax OPT- $x$  to OPT- $x$ -Relaxed by letting  $x_j$ 's to be real numbers, in order to make the optimization problem easier to be analyzed.

$$\begin{aligned} \text{OPT-}x\text{-Relaxed: } \max_{x'_j s} \quad & Q = \sum_{k=0}^n p_k \hat{\alpha}\left(\sum_{j=1}^k x_j\right), \\ \text{s.t.} \quad & \sum_{j=1}^n \frac{x_j}{j} \leq l, \\ & \sum_{j=1}^n x_j \leq d, \\ & x_j \geq 0, x_j \in \mathbb{R}, j = 1, 2, \dots, n. \end{aligned}$$

In the relaxed version OPT- $x$ -Relaxed, we replace  $\alpha(x)$  by its continuous version  $\hat{\alpha}(x)$ . Here  $\hat{\alpha}(x)$  can be any differentiable increasing function defined in  $(0, d]$  that satisfies:

$$\hat{\alpha}(x) = \alpha(x) \text{ when } x \in \mathbb{N}. \quad (5)$$

Let  $Q^*$  denote the optimal objective of OPT- $x$ -Relaxed. We have  $Q^* \geq H^* \geq J^*$ . That is,  $H^*$  is an upper bound of  $J$ .

### C. Finding An Upper Bound

We observe that if  $\hat{\alpha}(x)$  is a concave function, then the relaxed problem (OPT- $x$ -Relaxed) becomes a *convex optimization* problem, since the objective is concave and all constraints are linear. This class of problems can be efficiently solved using well-established methods such as gradient descent [17] or Newton's method [18]. Therefore, we can further relax OPT- $x$ -Relaxed by finding a concave  $\beta(x)$  making

$$\beta(x) \geq \hat{\alpha}(x), \quad 0 \leq x \leq d. \quad (6)$$

That is, we need to solve the following problem:

$$\begin{aligned} \text{OPT-UB: } \max_{x'_j s} \quad & U = \sum_{k=0}^n p_k \beta\left(\sum_{j=1}^k x_j\right), \\ \text{s.t.} \quad & \sum_{j=1}^n \frac{x_j}{j} \leq l, \\ & \sum_{j=1}^n x_j \leq d, \\ & x_j \geq 0, x_j \in \mathbb{R}, j = 1, 2, \dots, n. \end{aligned}$$

---

**Algorithm 1** Find an upper bound for  $J$ 

---

**Input:**  $\alpha(x)$ ,  $p_k$ 's,  $n$ ,  $l$ ,  $d$ .

**Output:** An upper bound  $U^*$ .

- 1: Find a concave  $\beta(x)$  that satisfies (6).
  - 2: Solve OPT-UB and get its optimal objective  $U^*$ .
- 

Let  $U^*$  denote the optimal objective of OPT-UB. We have  $U^* \geq J^*$ , i.e.,  $U^*$  is an upper bound of  $J$ . The procedure to find the upper bound is summarized in Algorithm 1.

Note that for many real-world datasets, the accuracy function  $\hat{\alpha}(x)$  is often approximately concave. As a result, the relaxation introduced in (6) is quite small. This observation will be further supported by our numerical results in Section V.

#### D. Theoretical Insights

Regarding OPT-UB, we have some theoretical insights that can help our scheduler design.

**Proposition 2** For any  $\beta(x)$ ,  $p_k$ 's,  $n$ ,  $l$ , and  $d$ , for the optimal solution  $x_1^*, x_2^*, \dots, x_n^*$  to OPT-UB, we have  $\sum_{j=1}^n \frac{x_j^*}{j} = l$ .

**A Proof Sketch** We provide a proof by contradiction. Suppose  $\sum_{j=1}^n \frac{x_j^*}{j} < l$ . We define  $q = \max\{i : x_i^* > 0\}$ . If  $q = 1$ , then we let  $x'_1 = l$ ,  $x'_2 = x'_3 = \dots = x'_n = 0$ . If  $q > 1$ , then we let  $x'_1 = x_1^* + \epsilon$ ,  $x'_q = x_q^* - \epsilon$  and  $x'_i = x_i^*$  for  $i \neq 1$  and  $i \neq q$ , where  $\epsilon$  is a small value. We can show that the solution  $x'_1, x'_2, \dots, x'_n$  has a greater objective than  $x_1^*, x_2^*, \dots, x_n^*$ , which contradicts to the optimality of  $x_1^*, x_2^*, \dots, x_n^*$ . ■

Proposition 2 implies that, under the optimal solution, the girth constraint (i.e., constraint (4)) is always tight.

For an integer  $v$ , we define that a solution  $x_1, x_2, \dots, x_n$  is a  $v$ -level solution if it satisfies  $\sum_{k=1}^n \mathbb{1}(x_k > 0) \leq v$ . In other words, a  $v$ -level solution uses at most  $v$  distinct priority levels in the PET scheme. Based on the description in [10], PET solutions with fewer priority levels are generally easier to construct.

We consider a special case of OPT-UB: the function  $\beta(x)$  is linear. We have the following result:

**Proposition 3** When  $\beta(\cdot)$  is linear, for any  $p_k$ 's,  $n$ ,  $l$ , and  $d$ , there exists a 2-level solution that is optimal to OPT-UB.

**A Proof Sketch** When  $\beta(\cdot)$  is linear, OPT-LB is an linear programming problem, and thus an optimal solution of it is a vertex of its feasible region. It can be shown that any vertex of the feasible region is a 2-level solution. ■

Let  $Q_{v\text{-level}}^*$  denote the optimal objective achieved by a  $v$ -level solution to OPT-UB. By Proposition 3, when  $\beta(\cdot)$  is linear, we have

$$Q_{2\text{-level}}^* = Q_{v\text{-level}}^* = Q^*, \text{ for } v \geq 2. \quad (7)$$

When  $\beta(\cdot)$  is not linear, 2-level solution is no longer optimal to OPT-UB. However, the following proposition provides a guarantee for the gap between 2-level solution and  $v$ -level solution.

**Proposition 4** For any concave  $\beta(\cdot)$ , we have

$$Q_{v\text{-level}}^* - Q_{2\text{-level}}^* \leq \frac{v-2}{3v}. \quad (8)$$

**A Proof Sketch** Define  $\delta = Q_{v\text{-level}}^* - Q_{2\text{-level}}^*$ . Consider the optimal  $v$ -level solution  $x_{s_1}, x_{s_2}, \dots, x_{s_v}$  with  $s_1 < s_2 < \dots < s_v$ . Define  $\beta_0 = \beta(0)$ . For  $i = 1, 2, \dots, v$ , define  $\beta_i = \beta(\sum_{j=1}^i x_{s_j})$ . Define  $q_i = p_{s_i}$ . We have  $Q_{v\text{-level}}^* = \sum_{i=0}^v q_i \beta_i$ . For  $i = 1, 2, \dots, v$ , define  $c_i = \beta_i - \beta_{i-1}$ . We have  $\sum_{i=1}^v c_i = \beta_v - \beta_0 \leq 1$ .

We consider  $\binom{2}{v}$  different 2-level solution with levels  $s_i$  and  $s_j$ , with  $x'_{s_i} = \sum_{k=1}^i x_{s_k}$  and  $x'_{s_j} = \sum_{k=i+1}^j x_{s_k}$ . It can be shown that the 2-level solution  $x'_{s_i}, x'_{s_j}$  is feasible to OPT-UB. Let  $Q_{i,j}$  denote its objective. For any  $i \neq j$ , we have

$$\begin{aligned} Q_{v\text{-level}}^* - Q_{i,j} &= \sum_{k=1}^{i-1} (q_k \sum_{w=1}^k c_k) + \sum_{k=i+1}^{j-1} (q_k \sum_{w=i+1}^k c_k) \\ &\quad + \sum_{k=j+1}^v (q_k \sum_{w=j+1}^k c_k). \end{aligned} \quad (9)$$

Since  $Q_{2\text{-level}}^* \geq Q_{i,j}$ , we have

$$\delta \leq \text{RHS of (9)}. \quad (10)$$

Multiplying both sides by  $q_i q_j$  and summing over all distinct  $(i, j)$  pairs yields:

$$\sum_{1 \leq i < j \leq n} q_i q_j \delta \leq \sum_{1 \leq i < j < k \leq v} q_i q_j q_k (c_1 + c_2 + \dots + c_n). \quad (11)$$

Recall that  $\sum_{1 \leq i \leq v} q_i \leq 1$  and  $\sum_{1 \leq i \leq v} c_i \leq 1$ . By Maclaurin's inequality [19], we have

$$\delta \leq \frac{\binom{3}{v}}{v \binom{2}{v}} = \frac{v-2}{3v}. \quad \blacksquare$$

Proposition 4 provides a theoretical guarantee that the loss in objective value incurred by using a 2-level PET solution instead of an optimal  $v$ -level solution is bounded and relatively small. For example, compared to a 3-level solution, the loss is at most  $\frac{1}{9}$ ; compared to an infinite-level solution, the loss is at most  $\frac{1}{3}$ .

#### IV. LOW-COMPLEXITY ALGORITHM DESIGN

Based on the theoretical insights from the previous section, we now propose an algorithm to construct PET. Our approach leverages the following insights from Propositions 1–4:

- The priority value of each word  $i$  is  $q_i/n$ , where  $q_i$  is an integer.
- The girth constraint (4) is tight under the optimal solution.
- The optimal solution involves at most two priority levels.

In [10], the authors present a method for constructing PET systems based on erasure-resilient codes. Specifically, an  $l$ -partition of  $m$  is defined as a sequence of positive integers  $\langle m_1, m_2, \dots, m_l \rangle$  such that

$$\sum_{j=1}^l m_l = m.$$

---

**Algorithm 2** Construct Optimal 2-Level PET Solution
 

---

**Input:**  $\alpha(x)$ ,  $p_k$ 's,  $n$ ,  $l$ ,  $d$ .

**Output:** PET parameters  $m$ ,  $\rho_1, \rho_2, \dots, \rho_m$ , PET solution  $\pi$ .

```

1: Set  $J^* = 0$ . For  $0 \leq i \leq j \leq n$ , set  $P_i^j = \sum_{k=i}^j p_k$ .
2: for  $m_1 = 1, 2, \dots, n-1$  do
3:   for  $m_2 = m_1 + 1, m_1 + 2, \dots, n$  do
4:     for  $l_1 = 1, 2, \dots, l$  do
5:       Set  $l_2 = l - l_1$ .
6:       if  $l_1 m_1 + l_2 m_2 \leq d$  then
7:         Set  $J = P_0^{m_1-1} \alpha(0) + P_{m_1}^{m_2-1} \alpha(l_1 m_1) +$ 
            $P_{m_2}^n \alpha(l_1 m_1 + l_2 m_2)$ .
8:         if  $J > J^*$  then
9:           Set  $l_1^* = l_1$ ,  $l_2^* = l_2$ ,  $m_1^* = m_1$ ,  $m_2^* = m_2$ ,
             and  $J^* = J$ .
10:        end if
11:      end if
12:    end for
13:  end for
14: end for
15: Set  $m = l_1^* m_1^* + l_2^* m_2^*$ . Set  $\rho_i = \frac{m_1^*}{n}$  for  $1 \leq i \leq l_1^* m_1^*$ 
   and  $\rho_i = \frac{m_2^*}{n}$  for  $l_1^* m_1^* + 1 \leq i \leq m$ .
16: Use the procedure in [10] to construct a PET solution  $\pi$ 
   with  $m$  and  $\rho_1, \rho_2, \dots, \rho_m$ .
```

---

Lemma 4.1 in [10] states that, given a number of packets  $n$  and an  $l$ -partition  $\langle m_1, m_2, \dots, m_l \rangle$  of  $m$ , one can construct a PET system consisting of  $n$  packets (each containing  $l$  words) such that all words in block  $j$  have a priority value of  $\frac{m_j}{n}$ . This PET system is implemented using erasure-resilient codes, and the full construction details can be found in [10].

We now return to our algorithm design. The goal is to construct an optimal PET system with at most two priority levels. Since the search space is relatively small under this constraint, we can afford to enumerate all feasible two-priority-level solutions and select the one that yields the highest objective value. The detailed procedure is presented in Algorithm 2. It is straightforward to verify that Algorithm 2 returns the optimal PET solution among all two-priority-level configurations.

Recall that when  $\beta(\cdot)$  is not linear, there might be a performance gap between a 2-level PET solution and a higher one. Other than the theoretical guarantee provided by Proposition 4, we have two more reasons for not considering more than two priority levels in our algorithm design.

- First, increasing the number of priority levels significantly increases the algorithm's time complexity. For instance, extending the strategy used in Algorithm 2 to search over all  $k$ -priority-level solutions (with  $k > 2$ ) results in a time complexity of  $O(n^k l^{k-1})$ . As  $k$  grows, this complexity becomes infeasible for meeting the stringent real-time constraints imposed by 5G systems.
- Second, and more importantly, our numerical results indicate that incorporating more than two priority levels yields only marginal improvements in the objective value  $J$ . This

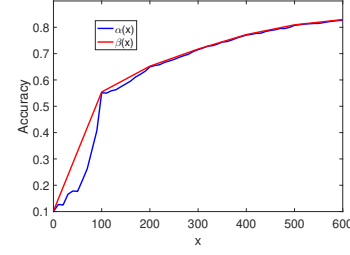


Figure 4: Accuracy function  $\alpha(x)$  under an NN-based compressor for CIFAR-10 and its concave approximation  $\beta(x)$ .

observation suggests that the added computational cost is not justified by the performance gain.

As a result, a low-complexity two-priority-level solution offers a more practical and efficient trade-off, and is adopted in our algorithm design.

Algorithm 2 has a time complexity of  $\mathcal{O}(n^2 l)$ , which is polynomial and generally sufficient to meet the real-time constraints of 5G systems. In particular, the maximum code block size is  $l = 264$  words (assuming each word is a 32-bit float number), and typical  $n$  values are in the tens. Thus, the  $\mathcal{O}(n^2 l)$  complexity is acceptable. For larger  $n$ , GPU-based parallelization may significantly accelerate processing [20], [21]. We plan to implement our method on real 5G platforms to validate this point in our future work.

## V. NUMERICAL RESULTS

In this section, we evaluate the performance of our proposed algorithm through numerical simulations. We compare the following:

- Our proposed 2-level PET solution (Algorithm 2).
- Upper bound provided by Algorithm 1.
- A naive transmission strategy without PET, where only the first  $e(t)$  words are transmitted without encoding (see the left part of Fig. 3).

We evaluate the performance of our proposed algorithm under realistic 5G settings. We consider a single-codebook compressor and classifier neural network (NN) trained on the CIFAR-10 dataset [22], following the approach described in [6]. The compressor NN produces  $d = 600$  output neurons. The accuracy function  $\alpha(x)$  is shown by the blue curve in Fig. 4, obtained by sampling classification accuracies for different values of  $x$ . Notably,  $\alpha(0) = 0.1$  and  $\alpha(600) = 0.83$ . We use the concave function  $\beta(x)$  (shown by the red curve in Fig. 4) to find the lower bound by Algorithm 1.

For the 5G transmission setting, we assume LDPC Base Graph 2 is used, where each code block contains 3824 information bits [11]. In our setup, each "word" corresponds to a 32-bit floating-point number. Hence, each code block can carry at most 119 words, i.e.,  $l = 119$ .

Fig. 5 shows the expected classification accuracy for BLER  $p$  with  $n = 5$  code blocks. The results align with those from artificial settings: the 2-level PET solution achieves performance



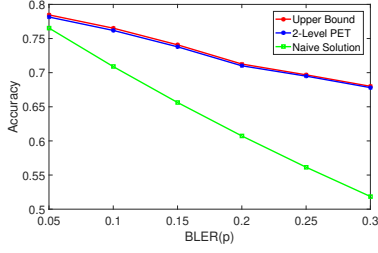


Figure 5: Expected classification accuracy for different BLER  $p$ 's for CIFAR-10 when  $n = 5$  and  $l = 119$ .

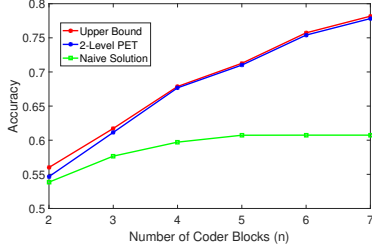


Figure 6: Expected classification accuracy for different  $n$ 's for CIFAR-10 when  $p = 0.2$  and  $l = 119$ .

very close to the upper bound, while significantly outperforming the naive baseline. For instance, at  $p = 0.3$ , the two-priority-level PET solution improves classification accuracy by 18% over the naive strategy—a substantial gain from a machine learning perspective. Fig. 6 shows the expected classification accuracy for various values of  $n$  (number of code blocks), with fixed  $p = 0.2$ . Again, the 2-level PET solution performs nearly as well as upper bound, and consistently outperforms the naive baseline in accuracy.

For our Matlab code on an Intel i7-12700 (2.10 GHz) CPU, the maximum wall-clock running time of Algorithm 2 for the simulations in Fig. 5 and Fig. 6 is 0.27 ms, which can satisfy the real-time constraint in 5G Numerologies 0 and 1. However, we acknowledge that our simulations were conducted with a maximum code block number  $n = 7$ , and for larger values of  $n$ , Algorithm 2 may exceed the real-time requirement. In such cases, a more efficient PET algorithm or GPU-based parallelization techniques may be necessary to meet timing constraints. We plan to explore these directions in future work.

## VI. CONCLUSION

In this paper, we developed an efficient source encoding framework for real-time edge classification tasks over 5G URLLC. Building on existing compression architectures with varying importance for different words, we integrated PET to enable priority-aware, progressive recovery of compressed image features. We formulated an optimization problem to determine the optimal PET configuration that maximizes expected classification accuracy under probabilistic code block loss. Along with the characterization of a theoretical bound on the performance of the optimal solution, a central finding of our work is that a two-priority-level PET scheme is sufficient to achieve near-optimal performance in many practical settings.

This insight enabled the development of a low-complexity algorithm suitable for deployment under the tight timing constraints of 5G. Our numerical experiments confirmed that the proposed 2-level PET approach significantly outperforms our baselines and achieves near-optimal performance.

## REFERENCES

- [1] W. Shi, J. Cao, Q. Zhang, Y. Li, and L. Xu, "Edge computing: Vision and challenges," *IEEE Internet of Things Journal*, vol. 3, no. 5, pp. 637–646, 2016.
- [2] T. Taleb, K. Samdanis, B. Mada, H. Flinck, S. Dutta, and D. Sabella, "On multi-access edge computing: A survey of the emerging 5g network edge cloud architecture and orchestration," *IEEE Communications Surveys & Tutorials*, vol. 19, no. 3, pp. 1657–1681, 2017.
- [3] N. Abbas, Y. Zhang, A. Taherkordi, and T. Skeie, "Mobile edge computing: A survey," *IEEE Internet of Things Journal*, vol. 5, no. 1, pp. 450–465, 2018.
- [4] Y. Matsubara, M. Levorato, and F. Restuccia, "Split computing and early exiting for deep learning applications: Survey and research challenges," *ACM Computing Surveys*, vol. 55, no. 5, pp. 1–30, 2022.
- [5] Y. Matsubara, S. Baidya, D. Callegaro, M. Levorato, and S. Singh, "Distilled split deep neural networks for edge-assisted real-time systems," in *Proceedings of the 2019 Workshop on Hot Topics in Video Analytics and Intelligent Edges*, 2019, pp. 21–26.
- [6] C. Li, P. Ju, A. Eryilmaz, and N. Shroff, "Efficient multi-dimensional compression for network-edge classification," in *Proceedings of the Twenty-fifth International Symposium on Theory, Algorithmic Foundations, and Protocol Design for Mobile Networks and Mobile Computing*, 2024, pp. 91–100.
- [7] 3rd Generation Partnership Project (3GPP), "3GPP TS 38.300: NR; NR and NG-RAN Overall Description," 3GPP, Tech. Rep. TS 38.300, March 2024, available at <https://www.3gpp.org/DynaReport/38300.htm>.
- [8] T. Richardson and S. Kudekar, "Design of low-density parity check codes for 5g new radio," *IEEE Communications Magazine*, vol. 56, no. 3, pp. 28–34, 2018.
- [9] H. Chen, R. Abbas, P. Cheng, M. Shirvanimoghaddam, W. Hardjawana, W. Bao, Y. Li, and B. Vucetic, "Ultra-reliable low latency cellular networks: Use cases, challenges and approaches," *IEEE Communications Magazine*, vol. 56, no. 12, pp. 119–125, 2018.
- [10] A. Albanese, J. Blomer, J. Edmonds, M. Luby, and M. Sudan, "Priority encoding transmission," *IEEE transactions on information theory*, vol. 42, no. 6, pp. 1737–1744, 1996.
- [11] 3rd Generation Partnership Project (3GPP), "3GPP TS 38.214: NR; Physical layer procedures for data," 3GPP, Tech. Rep. TS 38.214, March 2024, available at <https://www.3gpp.org/DynaReport/38214.htm>.
- [12] V. Saxena and J. Jaldén, "Bayesian link adaptation under a bler target," in *2020 IEEE 21st International Workshop on Signal Processing Advances in Wireless Communications (SPAWC)*. IEEE, 2020, pp. 1–5.
- [13] S. Park, R. C. Daniels, and R. W. Heath, "Optimizing the target error rate for link adaptation," in *2015 IEEE Global Communications Conference (GLOBECOM)*. IEEE, 2015, pp. 1–6.
- [14] M. Luby, "Lt codes," in *The 43rd Annual IEEE Symposium on Foundations of Computer Science, 2002. Proceedings*. IEEE Computer Society, 2002, pp. 271–271.
- [15] A. Shokrollahi, "Raptor codes," *IEEE transactions on information theory*, vol. 52, no. 6, pp. 2551–2567, 2006.
- [16] B. Masnick and J. Wolf, "On linear unequal error protection codes," *IEEE Transactions on Information Theory*, vol. 13, no. 4, pp. 600–607, 1967.
- [17] S. P. Boyd and L. Vandenberghe, *Convex optimization*. Cambridge university press, 2004.
- [18] J. Nocedal and S. J. Wright, *Numerical optimization*. Springer, 2006.
- [19] G. H. Hardy, J. E. Littlewood, and G. Pólya, *Inequalities*. Cambridge university press, 1952.
- [20] Y. Huang, S. Li, Y. T. Hou, and W. Lou, "Gpfc: A gpu-based design to achieve ~100  $\mu$ s scheduling for 5g nr," in *Proceedings of the 24th Annual International Conference on Mobile Computing and Networking*, 2018, pp. 207–222.
- [21] A. Kelkar and C. Dick, "Nvidia aerial gpu hosted ai-on-5g," in *2021 IEEE 4th 5G World Forum (5GWF)*. IEEE, 2021, pp. 64–69.
- [22] A. Krizhevsky, G. Hinton *et al.*, "Learning multiple layers of features from tiny images," 2009.